SCHOOL OF ENGINEERING

Diploma in Electronics & Computer Engineering (EGDF20)

Wireless Communication & Networking (EGE321)

Tutorial 2: Communication System Architecture

1) *Define what is multiple access methods?*

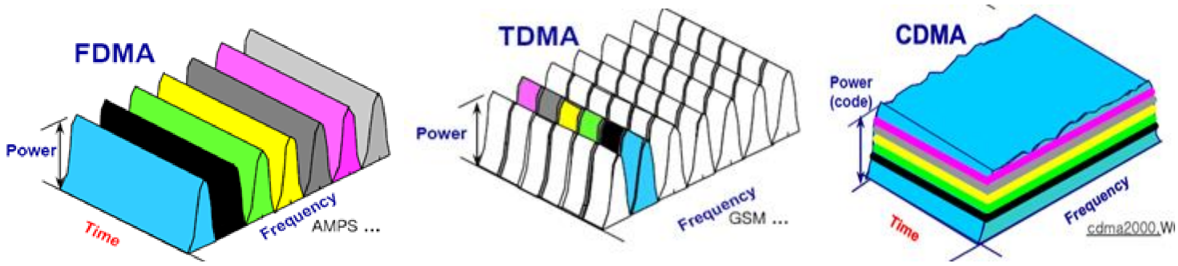
- Allow many users at the same time
- Share a finite amount of radio spectrum
- High performance
- Duplexing generally required

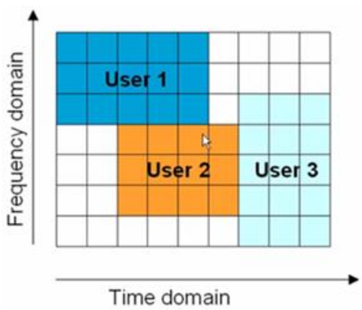
2) *What are the different types of multiple access methods?*

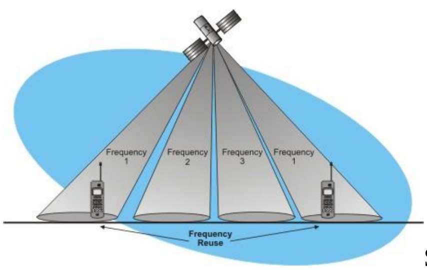
There are five basic access or multiplexing methods:

- frequency division multiple access (FDMA),
- time division multiple access (TDMA),
- code division multiple access (CDMA),
- orthogonal frequency division multiple access (OFDMA), and
- spatial division multiple access (SDMA).

3) Sketch the frequency vs time graph of the multiple access system. Explain briefly how the information is carried and distinguish by the multiple access system.







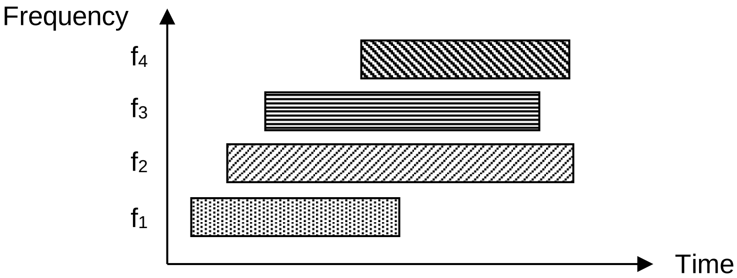
SDMA

4) *Explain, with the help of frequency-time diagrams, how the following multiple access schemes can be implemented :*

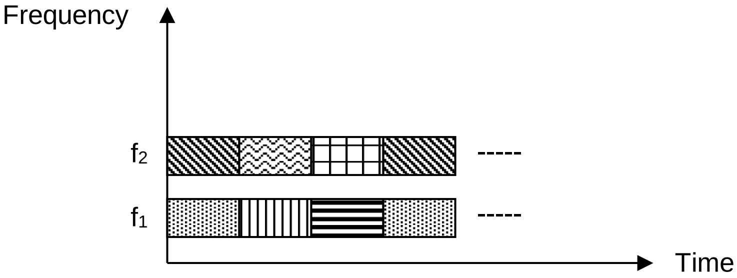
(i) Four users using FDMA

(ii) Six users using a combination of FDMA (2 frequencies) and TDMA

i) FDMA with four users



(ii) FDMA/TDMA for six users



5) *What is the difference between Multiple Access & Multiplexing?*

Multiplexing

Multiplexing is the process of transmitting several messages simultaneously on the same circuit or channel.

Multiple Access

Multiple Access are techniques that allow spectrum and power to be shared efficiently among multiple users. In multiple access, more than one simple signal can thus be transmitted as part of a single complex signal and separated out at the receiving end. This is not possible in multiplexing

6) *What is the frequency spectrum allocated for FM Radio Broadcasting? Considering each channel occupies 200 kHz, what is the maximum no. of channels available for broadcasting?*

For FM broadcasting: 88 – 108 MHz
Total frequency range available: 108-88 = 20 MHz
Each channel occupies 200 kHz

$\therefore$ Max. no. of channels available = 20MHz / 200 kHz = 100

7) *State the Shannon-Hartley Capacity Theorem and explain how the Shannon-Hartley Capacity can be increased.*

Shannon-Hartley Capacity Theorem: $C = B \log_2(1+S/N)$ bps

BW and S/N is proportional to C. Hence C can be increase by increasing the BW or S/N.

8) *A Class 1 telephone line has a flat bandwidth of 300Hz to 3400Hz and a minimum signal to noise ratio of 40dB. It is required to send data over the telephone line, what is the maximum possible data rate?*

$C = B\log_2(1+S/N)$ bps

$C = (3400-300)\log_2(1+10^{40/10}) = 41.192$kbps

9) *It is very critical to maintain the **Q**uality **o**f **S**ervice (QoS) in wireless communication system. State the three monitoring factors of QoS.*

Latency, Reliability, Security